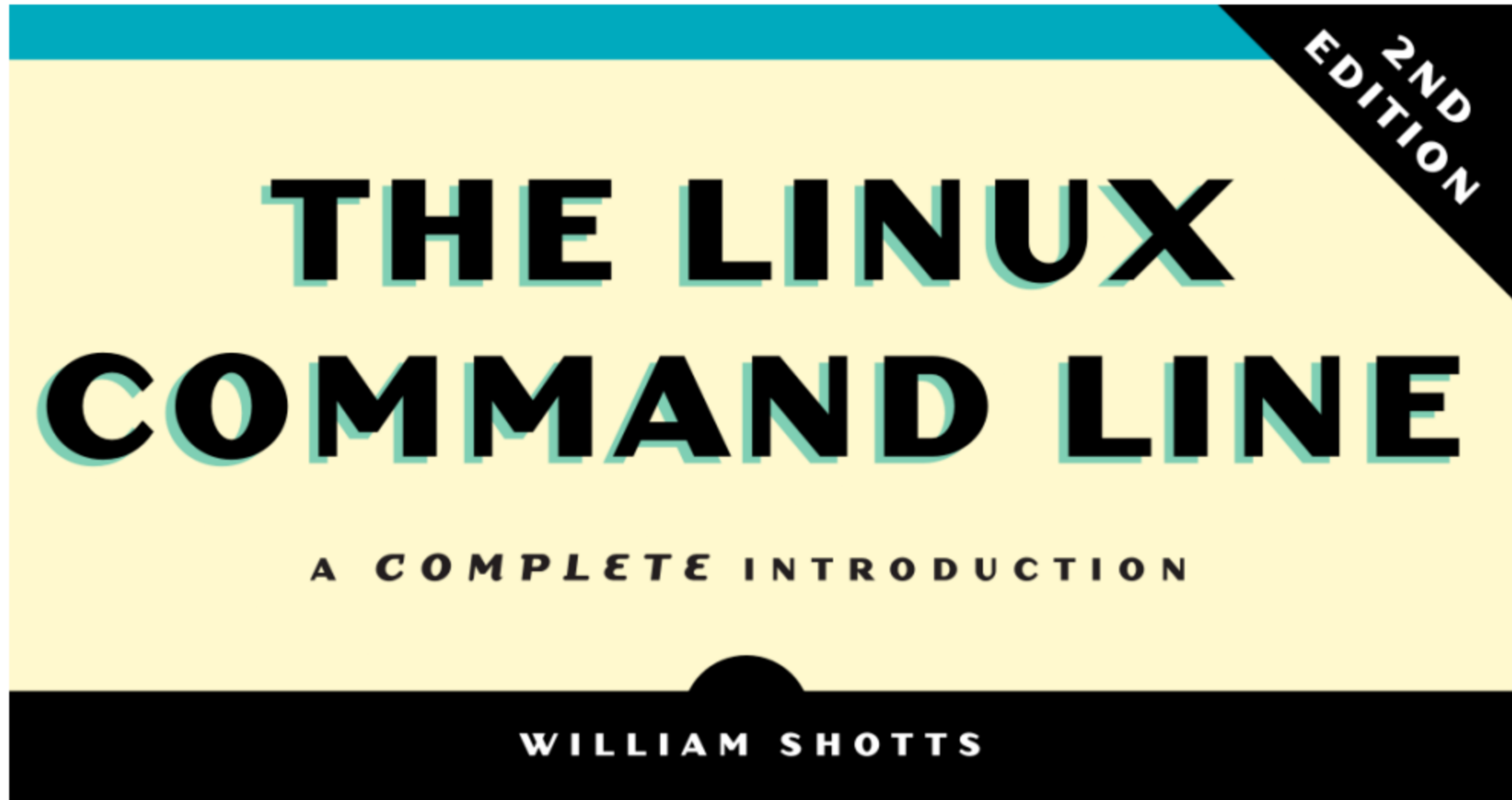


Ubuntu Operating System



<https://linuxcommand.org/tlcl.php>

The system can be used

- ▶ By multiple users (Multiuser)

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Assuming that you have logged into a Ubuntu System ...

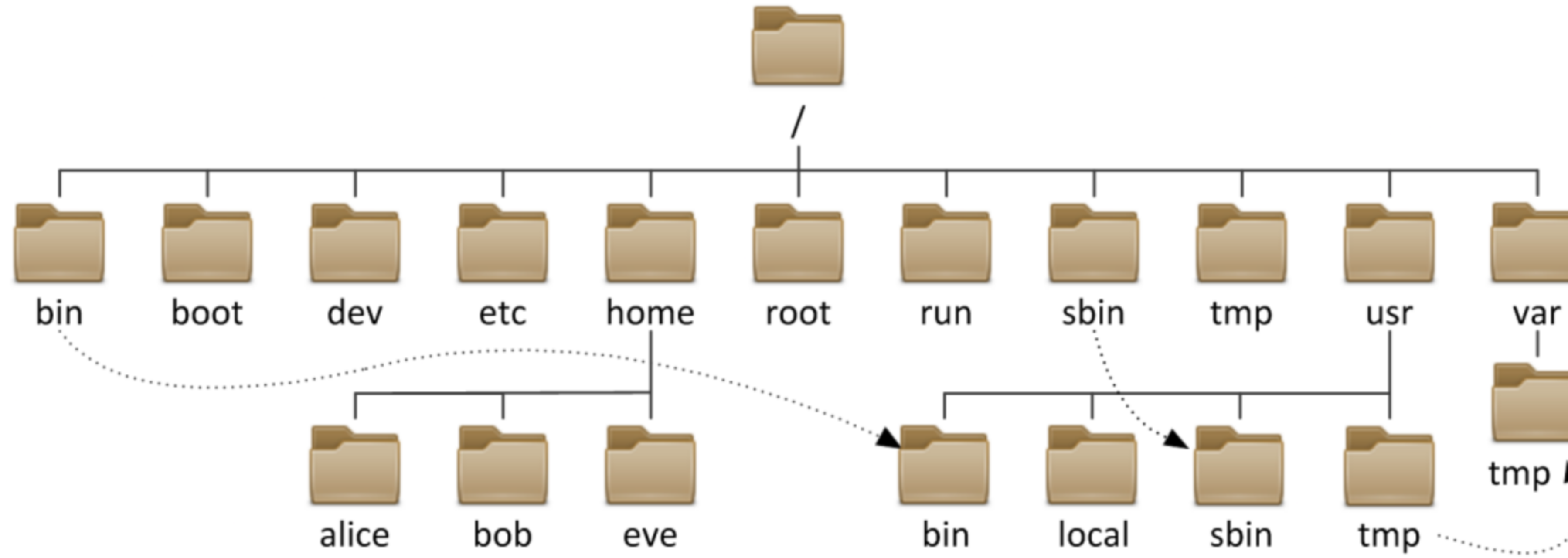
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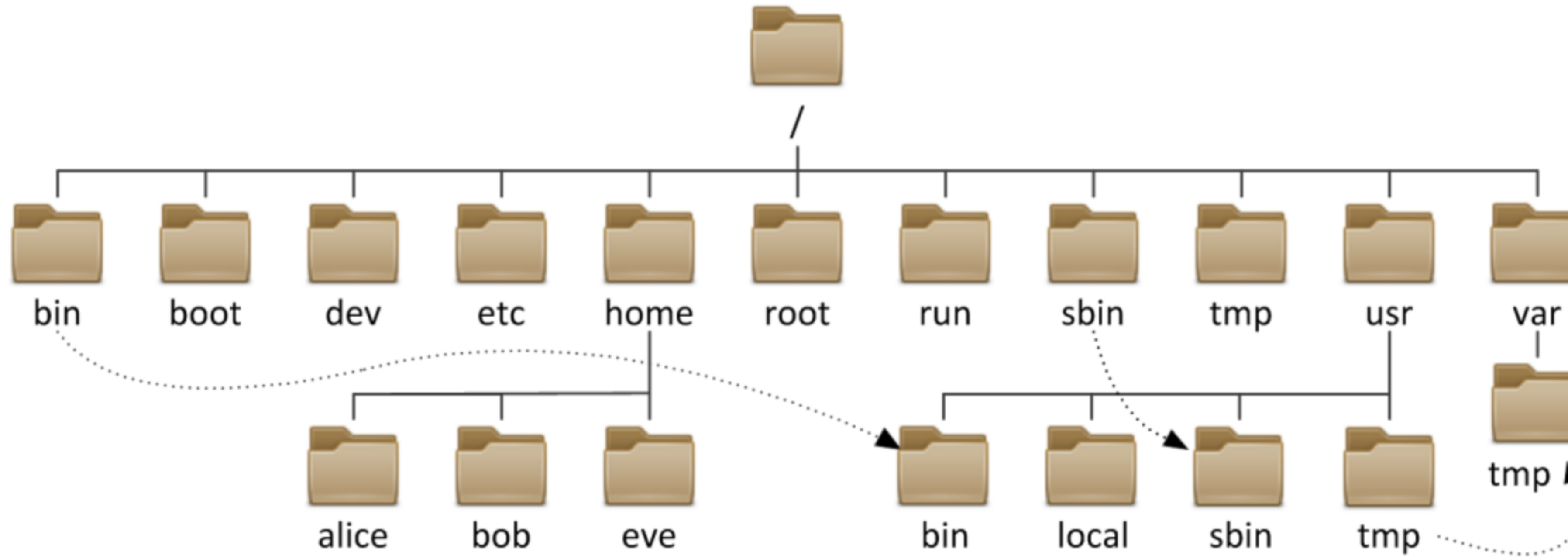
Assuming that you have logged into a Ubuntu System ...

- ▶ You will see desktop similar to Windows

Rooted-Tree File System

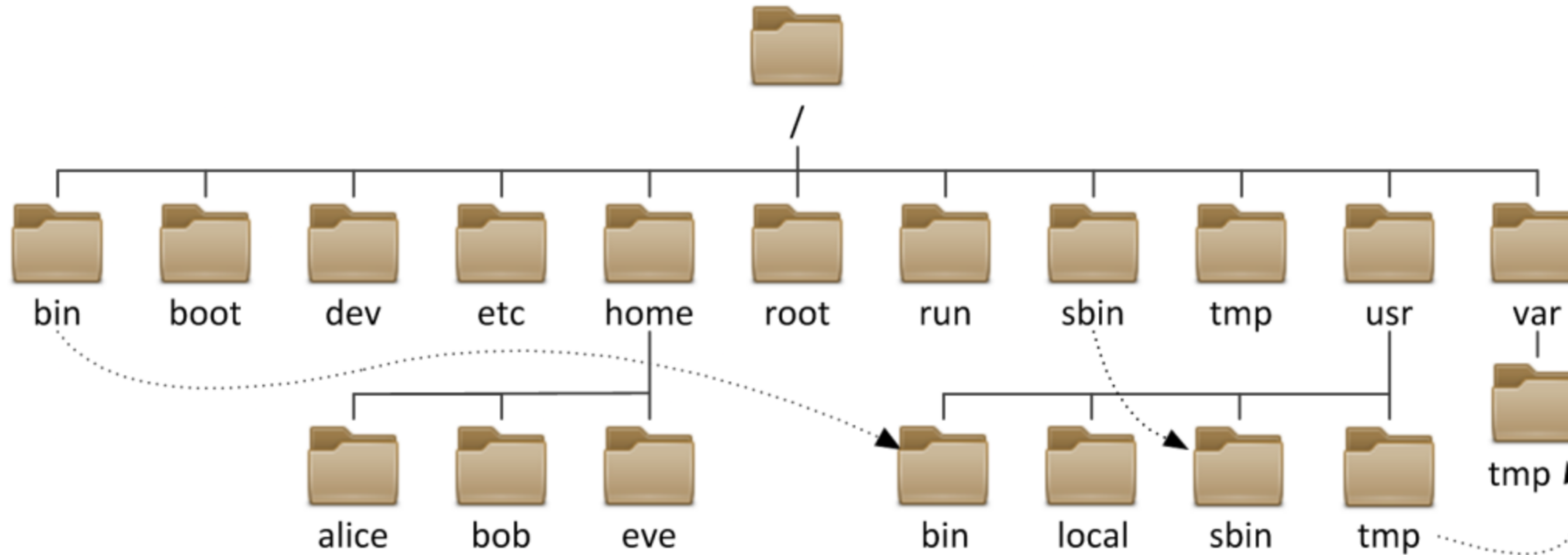


Rooted-Tree File System



As a user, you have a HOME folder (eg. /home/kavita)

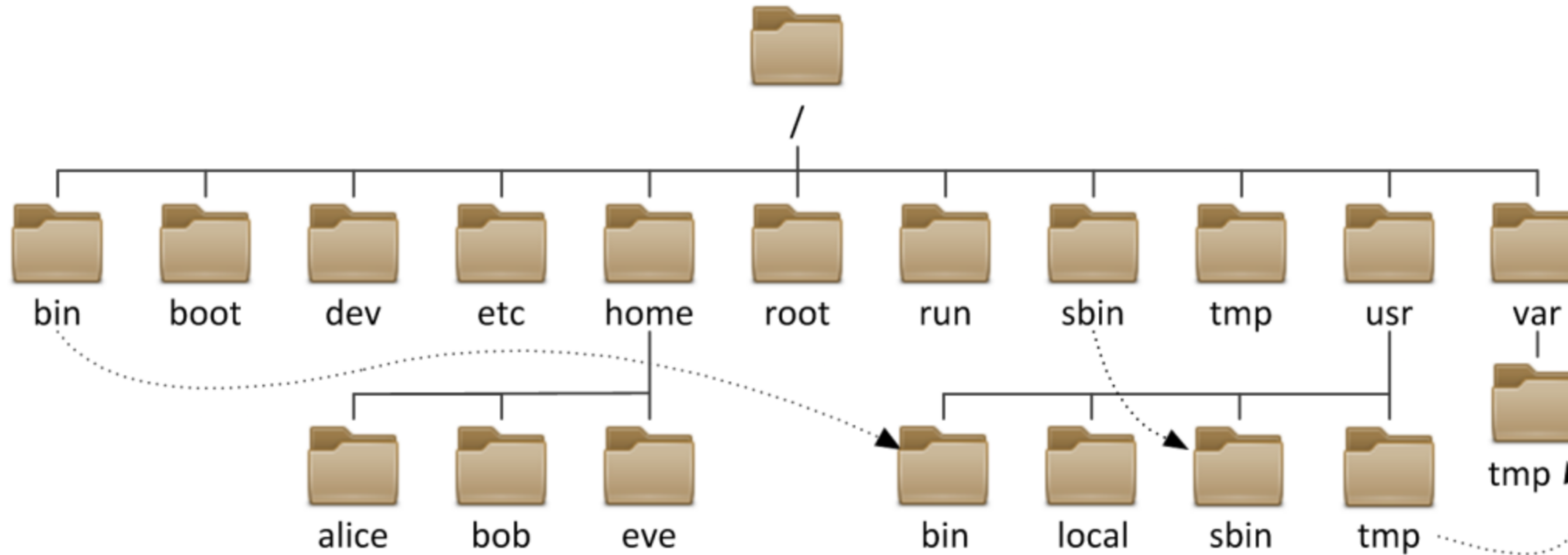
Rooted-Tree File System



As a user, you have a HOME folder (eg. /home/kavita)

Open a **Terminal** from the menu

Rooted-Tree File System



As a user, you have a HOME folder (eg. /home/kavita)

Open a **Terminal** from the menu

Change your password – Two ways –

(a) GUI

(b) Terminal - command `passwd`

HOME folder and Present Working Directory

When you start `Terminal` you are in HOME folder.

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Check the Present Working Directory

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$ pwd  
/home/...
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HOME folder and Present Working Directory

When you start **Terminal** you are in HOME folder.

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/home/...
```

Change directory to root

```
$ cd /
```

Nothing will be displayed. You can verify using `pwd`.

File Management with System Utilities

List Files

List files in the current folder.

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```
$ ls  
bin boot ...
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Option `-l` – show details

```
$ ls -l  
total 2.1G  
drwxrwxrwt 24 root root 4.0K Oct 8 15:58 tmp  
...
```

Shell Commands ...

Change directory to etc folder:

Shell Commands ...

Change directory to etc folder:

```
$ cd etc
```

Shell Commands ...

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List files

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...
```

Shell Commands ...

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List files

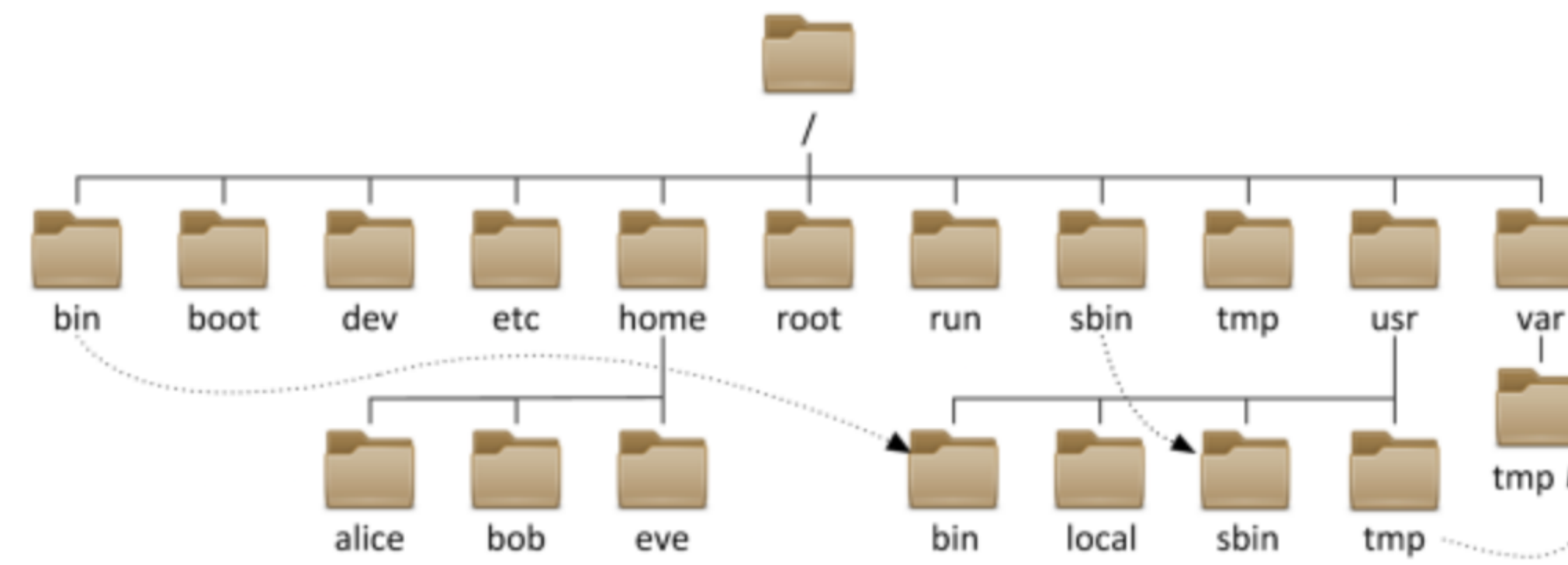
```
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...
```

Go to HOME folder

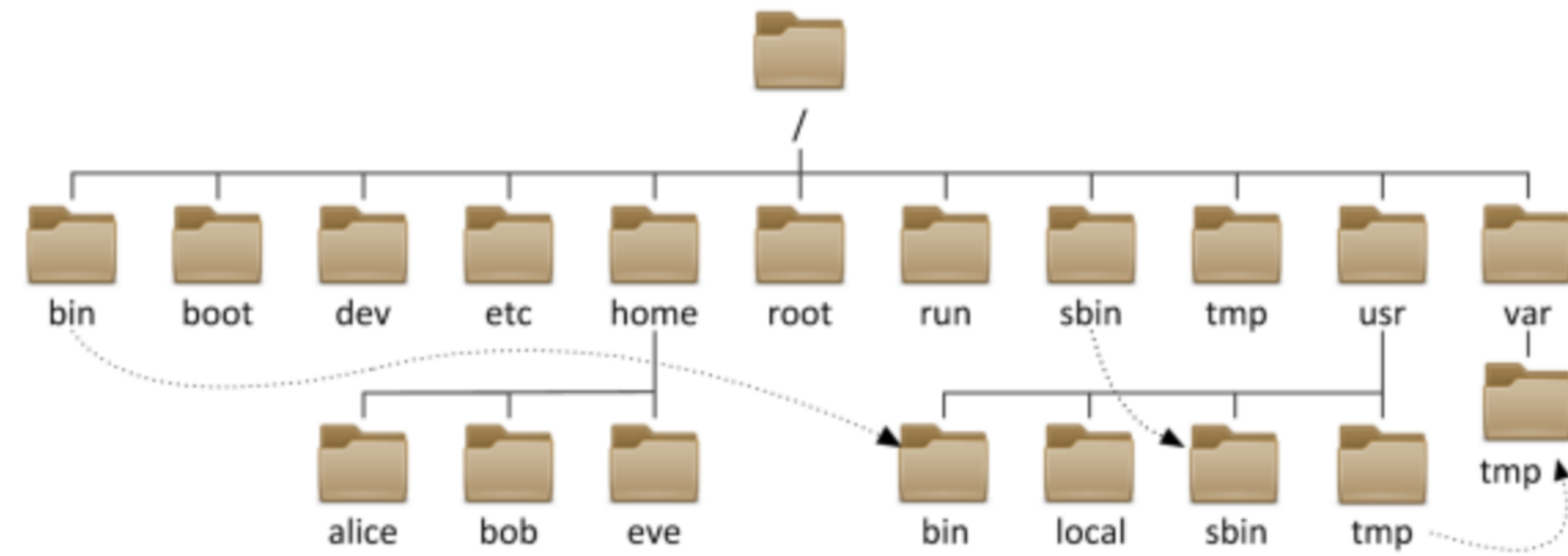
```
$ cd
```

Paths

Paths

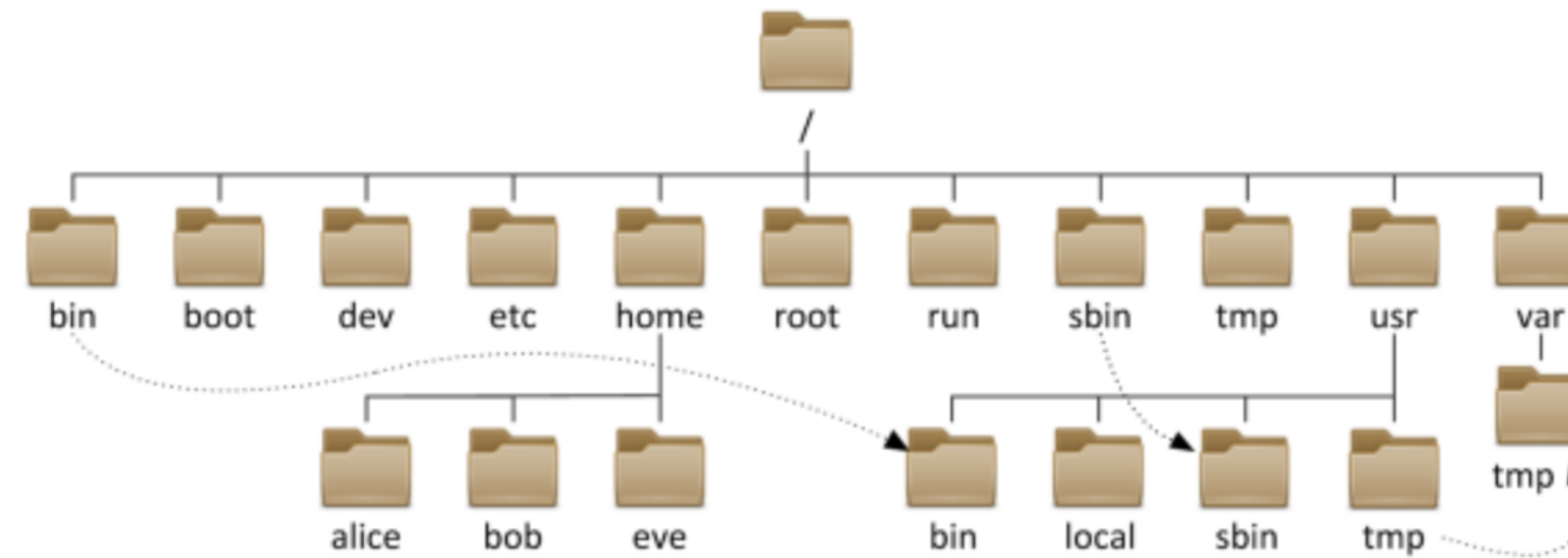


Paths



Relative path to a file or folder

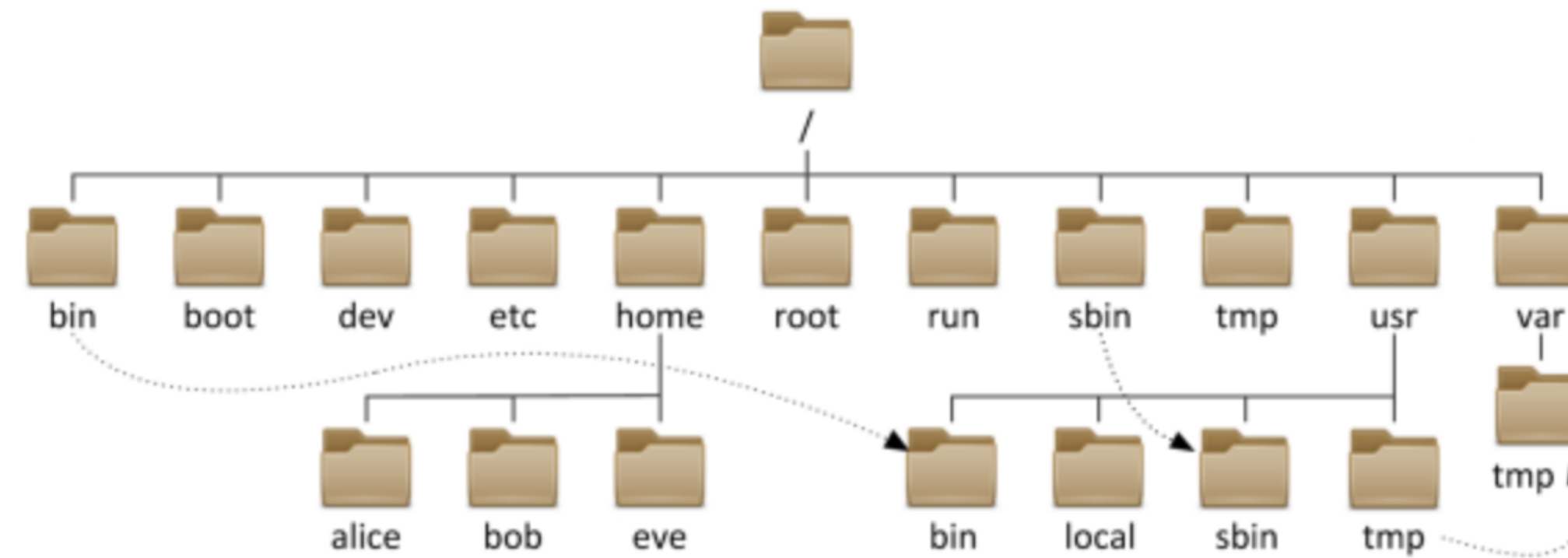
Paths



Relative path to a file or folder

```
$ ls ../../etc
```

Paths



Relative path to a file or folder

```
$ ls ../../etc
```

Absolute path

```
$ ls /etc
```


File Commands ...

```
$ mkdir ds3294
```

Check your current folder and confirm

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Other Commands:

▶ touch, cat, less # File creation/viewing

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How to find details of a command? `man`

System Info Commands

System Info Commands

CPU Information

System Info Commands

CPU Information

```
$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
...
Vendor ID: GenuineIntel
Model name: Intel(R) Core(TM) Ultra 5 125U
Thread(s) per core: 2
Core(s) per socket: 12
Socket(s): 1
...
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Get amount of free and used memory

System Info Commands

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$ free -h
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System Information ...

Kernel Info

```
$ uname -a
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Full hardware summary

```
$ lshw -short
```

System Information ...

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Full hardware summary

```
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```

Block devices

```
$ lsblk
```

Process Information

Get live information about current processes

```
$ top
```

Process Information

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```
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```

Process Information (Snapshot)

```
$ ps -af
```

Explore System Boot Process

Kernel buffer messages)

```
$ dmesg | less
```

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System boot logs

```
$ journalctl -b
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Currently Running Services

```
$ systemctl list-units --type=service --state=running
```